SIMATS ENGINEERING

ASSESSMENT TOOL 2 – Concept Mapping

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 **Total Marks: 100**

Annexure A

CONCEPT MAPPING

Code of Conduct:

I s.sri mathi(192424095) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Sri Mathi

Annexure A

CONCEPT MAPPING

1. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:

   a. Show differences and similarities

      i. Map how each affects the Data structure and Security level

      ii. Analyze which is more vulnerable to attacks and why

2. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:

   a. Show relationships between these concepts

   b. Analyze how each contributes to secure communication

   c. Include one real-world application for each.

CO1 - Assessment 2 - Concept Mapping

Classical Encryption Schemes

Substitution Techniques & Transposition

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Substitution Techniques

Replace each plaintext char with another character

Eg: Caesar Cipher, Monoalphabetic Polyalphabetic

Similarities

- Both are classical encryption
- Both require a key
- Both aim to hide the original message
- Both can be broken

Transposition Techniques

Keep the same characters but change their order in the ciphertext using a key or pattern.

Eg: Rail Fence Cipher, Columnar Transposition

Effect on Data Structure

- Changes the actual characters in the message
- Same length, but content looks different
- Structure remains, but with different symbols.

Effect on Security Level

Moderately secure

Vulnerable to frequency analysis

Better if polyalphabetic methods are used

Effect on Data Structure

Does not change the characters

Only changes the order

Same length, same set of characters

Which is more vulnerable to attacks and Why?

Substitution techniques are generally more vulnerable because they change the actual characters, making frequency analysis and pattern recognition easier. Transposition techniques are slightly safer as they keep the same characters, so attackers need more advanced methods to break them.

STEGANOGRAPHY
(Hiding information in plain sight)

Both are forms of steganography

Data Hiding
Hides secret data within a cover medium (image, audio video, text) so that it is not noticible to others.

Contribution to Secure Communications
Keeps the existence of the message secret
Prevents unauthorized access or detection

Real world Application
e.g., Sending confidential data in an image (like a photo sent through messaging apps)

Watermarking
Embeds a visible or invisible mark (eg., ownership, authentication) into a host data (image, audio, video, etc).

Watermarking focuses on long term survival

Contribution to Secure Communication
Proves the source / ownership of data
Helps detect tampering or unauthorized use

Real world Application
eg: Copyright protection for media content (images, videos, audio).

Survivability
The ability of the hidden information or watermark to remain intact and usable even after common manipulations (eg: compression, noise)

Contribution to Secure Communication
Maintains the effectiveness of the hidden data over time
Reduces the risk of data loss or corruption

Real world Application
eg: Robust watermarking in digital videos (surviving compression, resizing, slight edits).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation